

TrueMoist Phase 3C Architecture Research Report

Executive summary

Because the original topic was unspecified, I am making one explicit assumption: the intended research target is **TrueMoist Phase 3C architecture definition**, not a generic soil-moisture survey. That is the most plausible reading of the uploaded material, because the active SOP is specifically titled “**Phase 3C — TrueMoist Engineering Architecture Definition**”, the Engineering Design Review fixes TrueMoist’s mission and minimum demonstrable success, and the decision register says Phase 3C still needs a single final architecture rather than preserved forks. [filecite\turn0file10](#) [filecite\turn0file3](#) [filecite\turn0file12](#)

The strongest conclusion from the project record and the last five years of literature is that **TrueMoist should not try to invent a new probe first**. The most defensible MVP is a **software-first compensation architecture** built around an **off-the-shelf analog capacitive probe**, a **soil-temperature sensor**, an **external ADC**, **offline-trained but locally executed regression**, and **controlled salinity as a calibration variable rather than live in-soil EC sensing**. That aligns with the project’s fixed constraints: local processing, one homogeneous soil for the MVP, explainability, affordability, and physical gravimetric ground truth. It also matches the project’s Phase 2 guidance to preserve innovation on the **algorithmic layer** and avoid custom probe hardware where the patent fence is strongest. [filecite\turn0file10](#) [filecite\turn0file9](#) [filecite\turn0file12](#)

My selected architecture is therefore:

ESP32-WROOM-32 + ADS1115 + one low-cost analog capacitive probe + one waterproof DS18B20 + offline-trained ridge-regularized second-order regression executed locally + USB serial logging + gravimetric validation at 105 ± 5 °C.

The key strategic choice is to **represent salinity rigorously during calibration and disturbance testing** instead of prematurely integrating a direct soil-EC channel into the MVP. That choice is justified because soil salinity, pore-water conductivity, and bulk/apparent soil EC are related but not interchangeable; low-frequency dielectric sensing is itself sensitive to conductivity and temperature; and EC electrodes require AC excitation to avoid polarization and electrolysis. In other words, a “live soil EC” channel risks becoming a second sensor-development project before the first compensation loop is proven. ¹

Recent work supports this direction. A 2024 comparison found that adding an **ADS1115** improved the maximum RMSE of low-cost capacitive sensors from **3.78% to 1.87% v/v** in medium-textured soil, while a 2025 field study showed that low-cost capacitive sensors can become practically useful after **calibration plus temperature compensation**. At the same time, a 2025 low-cost dual-frequency moisture-and-EC prototype still reported errors above 15% without stronger calibration discipline, which is a useful warning against assuming that more complex instrumentation automatically removes the calibration burden. ²

On cost, the selected procurement path remains comfortably inside the project envelope. Using Indian retail listings for currently available development boards and modules, a practical procurement total is about ₹1,600–₹2,800 if lab oven access is available, and about ₹2,000–₹3,300 if the team must also buy a 0.01 g pocket scale. That keeps the preferred $\leq$ ₹3,000 target realistic while leaving the ₹5,000 ceiling intact. [filecite\turn0file10](#) 3

Scope, assumptions, and method

TrueMoist is not supposed to become a generic irrigation controller, a lab-grade universal probe, or a cloud platform. The project documents define it much more narrowly: an **embedded intelligent sensing system** whose contribution is **local compensation** of low-cost moisture-sensor error caused by temperature, salinity/EC influence, drift, and repeatability limits. The MVP only has to prove that compensated estimates are **more stable or closer to gravimetric reference** than uncompensated output under controlled disturbances; it does **not** need to prove universality across all soils, crops, or seasons. [filecite\turn0file10](#) [filecite\turn0file3](#)

The project’s frozen constraints are unusually clear and matter directly to architecture selection. Phase 3C demands one coherent buildable architecture, not option preservation. The budget is **preferred $\leq$ ₹3,000, absolute maximum ₹5,000**. The timeline is about **two months**. Processing must be **local** with **no continuous cloud dependency**. The MVP should use **one homogeneous soil type, physical ground truth, and explainable, reproducible methods**. Phase 3B also records that TrueMoist already had the portfolio’s strongest feasibility consensus, with the only major unresolved hardware issue being whether an external ADC is necessary. [filecite\turn0file10](#) [filecite\turn0file12](#) [filecite\turn0file8](#) [filecite\turn0file9](#)

I treated the evidence in four layers. First, I treated the project’s uploaded documents as binding constraints. Second, I used peer-reviewed and primary technical literature from roughly the last five years to understand what modern low-cost moisture compensation actually looks like in practice. Third, I used official component documentation from Espressif, TI, Analog Devices, Bosch, DFRobot, and Analog Devices/Maxim for hardware selection. Fourth, I used Indian retailer listings only for current **price-range checking**, not for sensor physics. [filecite\turn0file10](#) 4

Source class	Main examples	How it was used
Project authority	EDR, Phase 3B synthesis, Phase 3C TrueMoist SOP, decision register	Fixed constraints, MVP success definition, allowed architecture space. filecite\turn0file3 filecite\turn0file9 filecite\turn0file10 filecite\turn0file12
Peer-reviewed research	2022–2025 papers on temperature compensation, calibration, data assimilation, and low-cost sensor evaluation	State of the art, recent developments, algorithm and validation implications. 5
Official component docs	Espressif, TI ADS1115, DFRobot SEN0193, DS18B20, ST MCU docs	Hardware capability, interfaces, ADC characteristics, memory headroom. 6

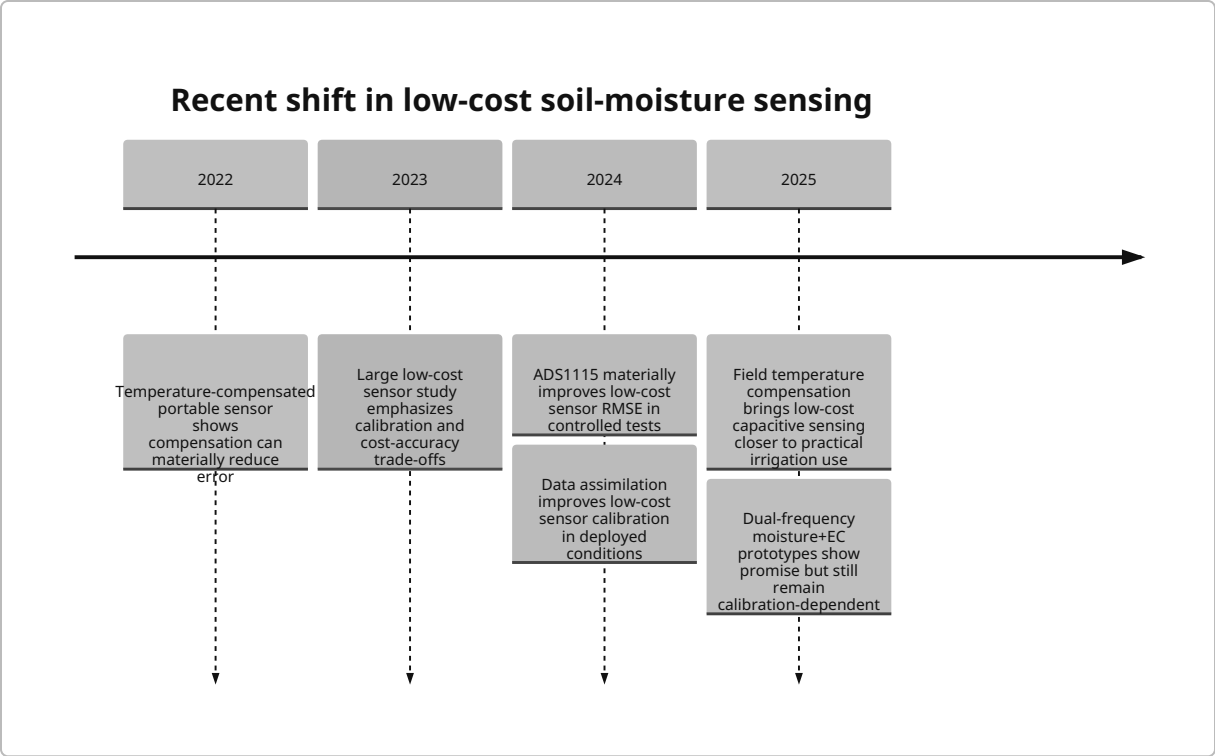
Source class	Main examples	How it was used
Price checks	Indian electronics retailers and lab suppliers	Current BOM range estimation only. ⁷

Background and recent developments

The core measurement problem is well understood. Low-cost capacitive moisture sensing is attractive because it is cheap and scalable, but the reading is not a pure moisture signal. Temperature and dissolved ions affect the dielectric response, especially at lower frequencies where **Maxwell-Wagner polarization** becomes important. That means a cheap sensor can be directionally useful yet still drift or misread when temperature changes, fertilizer or salts are present, or the same amount of water sits in different electrical conditions. ⁸

That physics matters because the project’s own Phase 3B synthesis already converged on the same idea: TrueMoist’s practical confounders are not abstract. The uploaded synthesis explicitly says the converged architecture is **capacitive probe + temperature + EC-informed compensation**, that calibration is the critical path, that gravimetric analysis is the required reference, and that the MVP should stay on **one soil type** rather than overclaim transfer. ⁹

The last five years of literature show a very clear trend: most meaningful progress has come less from “discovering a magical cheap probe” and more from **calibration, compensation, better instrumentation around the probe, and disciplined validation**. A 2022 study demonstrated temperature compensation on a portable soil-water sensor; a 2023 study emphasized the cost-versus-accuracy tradeoff of low-cost sensors; a 2024 study showed an inexpensive **ADS1115** can materially narrow the gap between low-cost and higher-end sensors; another 2024 paper showed calibration can be improved strongly with data assimilation; and a 2025 field study showed low-cost capacitive sensors can perform usefully after calibration and temperature correction. ⁹



The implication for TrueMoist is straightforward. The most defensible student-scale contribution is **not** to chase universal calibration-free moisture sensing. The decision register already warns that such claims remain unproven and that true multi-soil validation belongs to a longer, later sprint. For Phase 3C, the architecture should instead maximize **trustworthy local compensation on one homogeneous soil under controlled disturbances**, because that is both technically honest and fully aligned with the project’s fixed scope. [filecite\turn0file12](#) [filecite\turn0file10](#)

Architecture analysis and selected design

Architecture family comparison

The TrueMoist SOP defined four candidate architecture families. The right question is not “which sounds most sophisticated,” but “which family most cleanly proves the hypothesis under the team’s time, budget, and evidence constraints.” [filecite\turn0file10](#)

Family	What it does well	What breaks first	Evidence-based verdict
Architecture A minimal compensation demonstrator	Fastest to build; good as a baseline; controlled salinity compatible. filecite\turn0file10	Leaves the weakest unresolved point—ADC quality—inside the final design. 2024 evidence suggests that is an avoidable mistake. ¹⁰	Keep only as a Week 1 baseline , not as the final architecture.

Family	What it does well	What breaks first	Evidence-based verdict
Architecture B measurement-focused compensated sensor	Stronger measurement path and better analog discipline with an external ADC. filecite:turn0file10	Direct soil EC sensing adds electrode, polarization, excitation, and interpretation complexity; bulk EC and pore-water EC are not interchangeable. ¹¹	Use the external ADC idea , but do not make direct in-soil EC mandatory in the MVP.
Architecture C adaptive embedded calibration	Closest to the long-term research vision; compatible with future self-calibration work. filecite:turn0file10	Over-scoped for a two-month student MVP because it needs more data, drift history, and stronger safeguards against overfitting. ¹²	Defer to later phases after the fixed-model MVP is proven.
Architecture D offline training with embedded inference	Best fit to local processing, explainability, modest compute, and student workflow. It directly answers the SOP's requirement for one coherent build blueprint. filecite:turn0file10 filecite:turn0file12	Still requires disciplined dataset generation and gravimetric reference. ¹³	Selected family , with an external ADC and controlled salinity variable.

The competing viewpoints inside the uploaded project record map cleanly onto this table. The majority Phase 3B view supported a simple capacitive probe plus regression, while the minority view pushed either an **ADS1115** or a full **AD5933**-class impedance path. The best synthesis is not to average those positions, but to keep the **simple compensation architecture** while adopting the **cheap external ADC** that neutralizes the minority's strongest unrebutted concern. That is exactly the kind of one-path resolution Phase 3C is supposed to produce. [filecite:turn0file9](#) [filecite:turn0file10](#) [filecite:turn0file12](#)

Selected architecture

The selected architecture is a **D-dominant architecture with B-style signal hardening**:

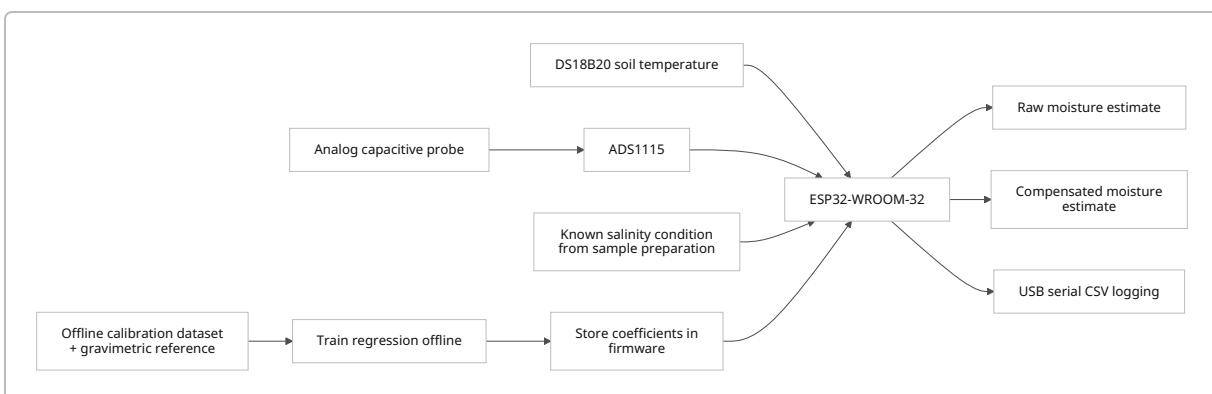
Design element	Selected choice	Rejected alternatives	Why this wins
Moisture sensor	One analog capacitive probe , preferably SEN0193-class or equivalent, with a second identical spare/profiling unit for characterization. ¹⁴	Resistive probe as primary sensor; custom probe; direct AD5933 probe front end. ¹⁵	Off-the-shelf, analog-accessible, cheap, and consistent with the project's instruction to stay on commodity hardware and keep IP ambition on the software layer. filecite:turn0file12

Design element	Selected choice	Rejected alternatives	Why this wins
Temperature input	Soil temperature near the moisture probe using a waterproof DS18B20 . ¹⁶	Ambient-only sensing; TMP36 analog; BME280. ¹⁷	DS18B20 is digital, simple, waterproof, and accurate enough for compensation. Ambient temperature can be logged later, but soil temperature is the more direct confounder for this MVP. ¹⁸
Salinity treatment	Controlled salinity as a calibration/disturbance variable , not a mandatory live in-soil sensor channel. If available, solution EC may be measured offline during preparation, but the runtime architecture must not depend on a field-robust soil EC subsystem. filecite turn0file10	Direct in-soil two-electrode EC; water-EC proxy used alone; destructive pore-water extraction in routine runtime. ¹¹	This satisfies the SOP's "measure or represent" requirement while avoiding a second difficult analog subsystem before the core compensation loop is proven.
ADC and analog front end	ADS1115 included by default . ¹⁹	ESP32 internal ADC only. ²⁰	The decision is justified by error budget, not by "more bits" alone: ESP32 ADC variation and nonlinearity are documented, and controlled testing shows the ADS1115 can materially improve low-cost sensor RMSE. ²¹
MCU	ESP32-WROOM-32 development board . ²²	ESP32-S3; STM32F103; STM32F401. ²³	Cheapest board with ample memory, broad library support, and strong alignment with the project's shared-platform guidance; once ADS1115 is added, the onboard ADC weakness stops dominating selection. filecite turn0file12
Model	Offline-trained ridge-regularized second-order regression , deployed as fixed coefficients on the MCU.	Pure linear-only model; random forest; TinyML; recursive/adaptive calibration.	More expressive than a plain linear fit, still interpretable, trivial to compute locally, and much safer than data-hungry adaptive methods in a one-soil two-month MVP. filecite turn0file10 ²⁴

Design element	Selected choice	Rejected alternatives	Why this wins
Interface and logging	USB serial + CSV logging , with optional simple status LED only.	App, OLED, cloud dashboard.	The project prioritizes evidence collection and interpretability over interface complexity. filecite[turn0file10]
Power	USB 5 V benchtop power , sensors and ADC on the regulated low-voltage rail.	Battery-powered MVP.	Battery behavior adds nothing to the proof of compensation and complicates measurement hygiene. The SOP explicitly tells Phase 3C to decide whether battery operation is necessary; here it is not. filecite[turn0file10]

The most controversial design choice is the salinity strategy, so it deserves plain language. I am **not** recommending that salinity be ignored. I am recommending that it be handled in the only way that is both rigorous and feasible for the MVP: as a **known experimental variable** used to create held-out disturbance conditions and model terms, rather than as a promise of reliable live in-soil EC estimation from a second improvised sensor. That keeps the architecture faithful to the project's definition of TrueMoist without overextending the prototype into a fresh electrochemistry problem. filecite[turn0file10] 25

The second controversial choice is the rejection of the **AD5933** path. The AD5933 is a capable impedance converter/network analyzer with programmable frequency sweep to 100 kHz and a much richer impedance workflow than a simple ADC path. But that sophistication is exactly why it is the wrong MVP default here: it adds excitation design, calibration, spectral interpretation, and extra integration work, while recent dual-frequency literature shows that richer measurement still does **not** eliminate the need for disciplined calibration. The right use of the AD5933 is as a later research upgrade if the simpler architecture proves the hypothesis and the team wants to extend toward deeper salinity/impedance modeling. 26



Validation, metrics, and bill of materials

Dataset and gravimetric reference

The MVP should validate against **gravimetric water content on a dry-mass basis** first, not against another low-cost sensor. Drying soil to constant mass at **105 ± 5 °C** is the standard reference method; volumetric water content can be derived later if bulk density is measured, but gravimetric water content is the cleaner first truth because it does not require introducing a second estimated variable. That is fully consistent with both the project's decision register and ISO soil-water standards. [filecite turn0file12](#) ¹³

A defensible Phase 3C dataset should be **grouped by physical sample preparation**, not by rows, because the SOP correctly warns against apparent accuracy created by repeated readings from the same soil state. The design below is a realistic compromise between statistical discipline and a two-month student schedule. [filecite turn0file10](#)

Dataset element	Recommended MVP setting	Why
Soil scope	One homogeneous, sieved local soil with texture documented once	The project docs explicitly prohibit overclaiming cross-soil generalization in the MVP. filecite turn0file10
Moisture levels	6 levels across the intended operating band, for example 5%, 10%, 15%, 20%, 25%, 30% gravimetric water content	Enough structure for nonlinearity without exploding workload.
Salinity levels	3 prepared solution bands , such as fresh / moderate / elevated salinity	Satisfies same-moisture different-salinity testing without requiring a live soil-EC subsystem. filecite turn0file10
Temperature levels	2 controlled soil temperatures for the core dataset, with a third optional stress level if time remains	Keeps workload realistic while still testing the confounder identified in the literature. ¹⁸
Replicates	3 independent soil preparations per condition	Separates physical repeatability from electronic read noise.
Reads per condition	10 sequential readings , aggregated by median and mean after stabilization	Soil is slow; repeated reads help estimate short-term repeatability.
Total physical conditions	108 = 6 moisture × 3 salinity × 2 temperature × 3 replicates	Large enough for a small regression model, still feasible in a semester prototype.
Split strategy	Grouped train/validation/test split by physical-condition batch	Prevents leakage from repeated reads of the same cup or state. filecite turn0file10

The gravimetric workflow should be boring and disciplined. Each soil batch should be mixed thoroughly, packed to a consistent fill mass and depth, probed using a fixed insertion guide, weighed wet, dried to constant mass at $105 \pm 5^\circ\text{C}$, cooled in a desiccator or sealed container, weighed dry, and then linked back to every electrical record from that batch. If lab oven access is uncertain, a microwave fallback exists in ASTM practice, but it is explicitly a faster substitute with lower reliability in some soil types, especially where salts or certain minerals are present. That makes microwave drying acceptable for intermediate checks, but **not** the preferred final reference for this project. ²⁷

I recommend **RMSE versus gravimetric water content** as the primary metric, because it is standard in the recent low-cost sensor literature and penalizes large misses. The secondary metrics should be **repeatability** under repeated insertion or repeated read conditions, and **disturbance sensitivity reduction**, which can be expressed as the percent error reduction of the compensated estimate versus the raw estimate under temperature and salinity changes. A practical success target for the MVP is a **20–30% reduction in grouped-test RMSE**, with this report recommending **25%** as the internal design target. That target is an engineering recommendation, not a published standard. It is chosen because the project only needs to show a meaningful, interpretable improvement over raw output, and recent literature indicates that materially better low-cost performance is achievable when compensation and instrumentation are handled carefully. ²⁸

Error budget, compute, and power

Before empirical characterization, the cleanest **qualitative pre-build error budget** is this: the biggest likely uncertainties are **soil packing and insertion consistency**, **gravimetric discipline**, **probe-to-probe variability**, **temperature gradients inside the sample**, and then only after that **ADC/reference quality**. That ordering matters. It means the ADS1115 is worth including because it removes one avoidable electronics penalty, but it also means the project will still fail if cups are packed inconsistently, soil evaporates during handling, or the dataset split leaks repeated states into both training and testing. ²⁹

The compute budget is very comfortable. ESP32 provides **520 KB SRAM** on chip, and common WROOM-class development boards offer **4 MB flash**. A fixed regression model with roughly 8–12 stored coefficients, a small moving-average buffer, and serial logging uses negligible memory and negligible compute by comparison. This is one reason to prefer a fixed, interpretable regression model over TinyML for the MVP: the constraint is not compute scarcity, but validation discipline and clarity. ²²

The power budget is similarly nonbinding for a benchtop MVP. The ADS1115 is a low-power ADC with typical power consumption around **0.3 mW**, and the DS18B20 was designed for low-component-count digital temperature sensing. Since the proof target is compensation quality rather than field endurance, **USB power** is the right architectural choice. It removes batteries, chargers, and brownout analysis from the critical path and makes analog repeatability easier to debug. ³⁰

Bill of materials

The BOM below separates **project-essential procurement** from **institutional equipment**. That distinction matters, because gravimetric truth depends much more on oven access than on any single electronic part. ³¹

Item	Suggested quantity	Current India range	Notes
ESP32 development board	1	₹399–₹549	Widely available 30-pin and 38-pin dev boards. ³¹
ADS1115 module	1	₹119–₹325	Cheap enough to include by default. ³²
Analog capacitive moisture probe	2	₹60–₹710 each	Low-cost clones are very cheap; branded SEN0193-class parts cost more. Two units help characterize sensor-to-sensor spread. ³³
Waterproof DS18B20 probe	1	₹64–₹72	Soil-temperature input. ³⁴
Perfboard, headers, resistors, capacitors, jumper wire	1 set	₹150–₹300	Engineering estimate; use soldered final analog path after early characterization.
Containers, labels, insertion guide materials	1 set	₹150–₹300	Engineering estimate.
Lab-grade NaCl for controlled salinity	1 pack	~₹342 / 500 g	Enough for many prepared conditions. ³⁵
Optional 0.01 g scale if lab scale unavailable	1	₹349–₹499	Only if institutional equipment is missing. ³⁶

Using the low-cost-to-midrange procurement path, electronics and consumables stay around **₹1,600–₹2,800** before optional scale purchase. Even with a purchased portable scale, the build remains near or under the preferred budget band, provided the team uses institutional oven access for final gravimetric validation.

³⁷

Implications, gaps, and prioritized recommendations

The practical implication of this architecture is that it gives TrueMoist a **clean, honest claim**: not “universal, calibration-free soil sensing,” but **demonstrable error reduction from local compensation on constrained hardware**. That is exactly the right level of ambition for the semester prototype. It also preserves the project’s longer-term innovation path, because the decision register already directs TrueMoist away from custom hardware IP and toward the algorithmic/software layer. The first MVP should therefore be judged by **trustworthiness improvement**, not by novelty theater. ³⁸

There are still real gaps. This architecture does **not** prove multi-soil generalization, long-term self-recalibration, or robust live in-soil EC estimation. The decision register rightly keeps those as later validation problems, and the literature supports that caution: multi-soil or calibration-free claims remain much harder

than single-soil compensated sensing. Likewise, the selected model may be technically successful while still not being patentable on its own; the SOP is explicit that ordinary polynomial regression on commodity sensors must not be mistaken for IP. [filecite turn0file12](#) [filecite turn0file10](#) 38

The highest-risk open question is not actually the regression model. It is **experimental discipline**. If soil packing varies, if evaporation is not controlled, if the probe is inserted at inconsistent depth or angle, or if the ground-truth pipeline is sloppy, the project can still produce attractive graphs and weak evidence. That is why the architecture should be implemented with a simple fixed insertion jig, a written sample-prep SOP, grouped data splits, and a rule that every test record remains traceable to one physical batch ID and one gravimetric measurement pair. [filecite turn0file10](#) 39

The prioritized next steps are therefore:

Priority	Action	Why it must happen first
Highest	Characterize one probe on ESP32 internal ADC versus ADS1115 in Week 1	This closes the only major unresolved instrumentation dispute quickly and cheaply. filecite turn0file9 10
High	Freeze the gravimetric SOP and physical sample-prep SOP before collecting the main dataset	True ground truth and repeatable packing matter more than model sophistication. filecite turn0file12 13
High	Collect grouped data on one soil across moisture, temperature, and salinity conditions	This is the minimum evidence needed to prove the compensation hypothesis honestly. filecite turn0file10
Medium	Fit and compare three models only: raw baseline, linear regression, ridge-regularized second-order regression	Keeps the model comparison interpretable and prevents over-scoping into novelty ML. filecite turn0file10
Medium	Deploy only the winning fixed model to the MCU and validate it on held-out physical batches	Local inference is the point of the concept; held-out physical batches are the real test. filecite turn0file10
Later	Treat live EC sensing, adaptive recalibration, and multi-soil transfer as post-MVP research extensions	These are valid next steps, but they are not required to prove the Phase 3C hypothesis. filecite turn0file12 40

The final recommendation is therefore unambiguous: **build TrueMoist as a one-soil, local-inference, ADS1115-backed capacitive compensation prototype using soil temperature and controlled salinity conditions, validate it gravimetrically, and judge success by grouped-test error reduction over the raw sensor**. That architecture is the best match to the uploaded project constraints, the best synthesis of the competing internal viewpoints, and the most evidence-backed way to produce a credible semester-scale demonstration. [filecite turn0file10](#) [filecite turn0file9](#) 41

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